

Auditing the 3rd-Place Solution to the Jigsaw Unintended Bias in Toxicity Classification Competition

Ian Tang and Nick Chen
DS-UA 202: Responsible Data Science, Spring 2026
Group 21

May 2, 2026

1 Introduction

This report audits the third-place finishing solution to the 2019 Jigsaw *Unintended Bias in Toxicity Classification* Kaggle competition, a publicly available toxicity-classification ensemble built by sakami0000 [sakami0000, 2019]. The dataset under audit is the Civil Comments corpus, released by Jigsaw alongside the competition [Borkan et al., 2019]; the system we recreate combines a fine-tuned **bert-large-uncased** with a five-fold LSTM-GRU model under Optuna-tuned blend weights (§4).

The audit takes a normative spine. The system implicitly enacts a particular theory of fairness through the metric it was trained against, the threshold at which it is deployed, the identity columns that metric evaluates, and the rater procedure that produced its training labels. Each of those is a value-laden choice the developers and the competition designers left implicit. Our task is to name the choices, test how well the implementation realizes the fairness ideals they encode, and surface the trade-offs the developers could not have avoided. With base rates that differ across identity subgroups, the Chouldechova impossibility result [Chouldechova, 2017] guarantees that no single fairness metric is neutral; the choice of which condition to privilege is therefore a stakeholder choice. Every numerical result reported in §5 is anchored to one of these normative claims.

The remainder of the report follows the rubric. §2 gives background on the dataset and the competition’s stated goals; §3 profiles the input and output and surfaces the toxicity-label measurement critique; §4 describes the recreated implementation and validates it against the competition’s bias-aware score; §5 reports outcomes across accuracy (§5.1), fairness (§5.2), and additional audit methods including counterfactual robustness (§5.3.1) and a critique of the competition metric itself (§5.3.2); §6 reflects on data appropriateness, deployability, and recommendations.

2 Background

The Civil Comments platform was a third-party comment-hosting service for English-language news sites between 2015 and 2017; on its shutdown, Jigsaw released the corpus of approximately 1.8M comments, with crowd-rated toxicity and identity annotations, as a public dataset [Borkan et al., 2019]. The 2019 Kaggle competition that accompanied the release framed two stated goals for participants: maximize the rank-quality of toxicity prediction overall, and minimize unintended bias against identity subgroups whose mention is correlated with toxic content in the training data.

The competition encoded the trade-off between these goals as a single bias-aware metric: an Overall AUC component weighted at 0.25 and a generalized power mean (with exponent $p = -5$) over per-identity bias AUCs weighted at 0.75, evaluated on nine of the twenty-four identity columns the dataset

annotates. The 25/75 weighting, the power-mean exponent $p = -5$, and the nine-of-twenty-four identity-inclusion cutoff each encode a normative claim that §5.3.2 unpacks; the competition’s defaults were not the only defensible ones.

The system audited here is reproduced from sakami0000’s publicly available code [sakami0000, 2019]. We audit the recreated BERT-large + LSTM-GRU ensemble, not the original five-component F.H.S.D.Y. ensemble, and are explicit about the recreation’s scope and its fairness implications in §4.1.

3 Input and Output

3.1 Civil Comments and identity-subgroup coverage

The audit runs on the private test set of the Civil Comments corpus ($n = 97,320$), released by Jigsaw alongside the 2019 Kaggle competition under audit [Borkan et al., 2019]. Civil Comments operated as a third-party comment-hosting platform for English-language news sites between 2015 and 2017; the public release contains approximately 1.8M comments collected when the platform shut down.

Each comment was annotated along two tracks. The toxicity track records a primary `toxicity` label and six subtype labels (`severe_toxicity`, `obscene`, `identity_attack`, `insult`, `threat`, `sexual_explicit`), each as the fraction of raters who applied the label. The identity track records twenty-four attributes spanning gender (4), sexual orientation (4), religion (7), race/ethnicity (5), and disability (4), each as the fraction of identity-annotators who marked the identity as referenced. Following Civil Comments convention, an identity is treated as “mentioned” when its column value is at least 0.5.

At threshold 0.5, the nine identities the bias-aware metric evaluates—`male`, `female`, `homosexual_gay_or_lesbian`, `christian`, `jewish`, `muslim`, `black`, `white`, `psychiatric_or_mental_illness`—together account for 11,003 mentions; the fifteen excluded identities (`transgender`, `heterosexual`, `bisexual`, `asian`, `latino`, `hindu`, `buddhist`, `atheist`, the four `other_*` category-residual columns, and every disability category other than `psychiatric_or_mental_illness`) account for only 807 in aggregate. All evaluated subgroups have $n \geq 238$ at this threshold; all excluded have $n \leq 217$. The selection appears sample-size driven, cutoff between those two values—revisited in §5.3.2. Per-group base rates among evaluated identities span $[0.0996, 0.3233]$ (`christian` to `black`); this range is binding for §5.2’s impossibility result.

3.2 Output: toxicity score and thresholding

The system produces seven floating-point outputs per comment, each in $[0, 1]$: a primary toxicity score $\hat{Y}$ and six subtype scores. Each score is trained to approximate the fraction of Civil Comments raters who applied the corresponding label. The competition metric and our §5.1 accuracy analysis treat $\hat{Y}$ as a continuous score and use AUC-style measures. Our §5.2 fairness analysis thresholds $\hat{Y}$ at $s_{HR} = 0.5$ to recover a binary moderation decision and report the resulting confusion-matrix metrics. We adopt 0.5 as a deliberately conventional choice—the default a deployed moderation pipeline would inherit from any binary classifier—so the threshold-side analysis is reproducible and a reader who prefers a different operating point can substitute it without re-running the model. Per the algebra of §5.2, the impossibility result holds at any non-trivial threshold given the base-rate spread reported above.

3.3 The toxicity label as a measurement choice

The toxicity label we audit against is itself the product of a measurement choice: it is the fraction of crowd-worker raters who applied the “toxic” label to a given comment. This formulation embeds three commitments that downstream fairness analysis inherits. First, it treats toxicity as an aggregable property of the text rather than a property of the rater–text dyad: a label value of 0.4 does not distinguish “a coherent minority of raters found this toxic” from “most raters disagreed about whether

it was toxic at all.” Second, training and evaluation both binarize the score around 0.5, so subgroups whose lived sense of what counts as a slur or a microaggression is a minority view among the rater pool are systematically under-counted as harmed parties. Third, the rater pool is not demographically balanced to the populations the labels are *about*. The audit that follows respects the label as given, but every fairness gap we report should be read modulo this upstream measurement choice—a different rater pool or a different aggregation rule could move some of the gaps materially.

4 Implementation and Validation

4.1 Recreation scope and deviations from the original

The original third-place solution was an F.H.S.D.Y. five-component ensemble combining several transformer- and recurrent-architecture variants together with fine-tuned GPT-2 language-model component(s). Recreating the full ensemble was infeasible within this project’s compute budget; we therefore reproduced two of its components—**bert-large-uncased** fine-tuned for the competition’s seven-way multi-task head, and a five-fold-averaged LSTM-GRU model with the same multi-task head structure—and combined them with Optuna-tuned blend weights. The fairness implication of this scoping is non-trivial. An ensemble whose structural diversity is reduced may encode a single component’s inductive bias more sharply than the original; conversely, biases that the original ensemble averaged out across components may surface more visibly here. We return to this point in §5.3.1 when comparing the BERT, LSTM, and ensemble bias profiles directly, and again in §6.4 when discussing what additional auditing the missing components would have warranted.

4.2 Architecture: BERT-large-uncased + LSTM-GRU multi-task ensemble

bert-large-uncased is the standard 24-layer, 1024-hidden, 16-attention-head transformer with $\sim 340\text{M}$ parameters; the recreation feeds its pooled output to seven linear heads (one toxicity head plus six subtype heads) and fine-tunes them jointly. The LSTM-GRU component is a recurrent sequence model with the same seven-head output structure, trained as a five-fold cross-validation ensemble whose per-fold scores are averaged at inference. Both models inherit Civil Comments’ tokenization and preprocessing as distributed by Jigsaw, with no project-specific cleaning beyond what the original recipe specifies. The two models are trained independently; only their per-comment scores are combined at inference time, which is what makes the blend weights below interpretable as relative trust placed on each component for the final score.

4.3 Validation: Optuna blend and competition-format reproduction

The blended ensemble’s per-comment toxicity score is the Optuna-tuned weighted mean of the BERT and LSTM scores. The optimizer returned blend weights of (BERT : 0.993, LSTM : 0.007) on the validation fold, meaning that under the competition’s bias-aware loss the LSTM contributes negligibly when BERT is available. We retain the LSTM in the audit nonetheless, because its standalone bias profile differs from BERT’s in ways that are analytically informative even when its blend weight is small (§5.3.1).

On the private test set ($n = 97,320$), the recreated ensemble reproduces the competition’s bias-aware score at 0.94464, with component AUCs of 0.97368 (Overall), 0.90183 (mean Subgroup), 0.94762 (mean BPSN), and 0.95544 (mean BNSP). These numbers anchor the rest of the audit: subsequent fairness gaps are read against a model whose competition-format performance is close enough to the third-place submission to support an audit of the recreated system, not the original five-component ensemble.

5 Outcomes

5.1 Accuracy across subpopulations

Per-identity AUC disaggregation operationalizes a discriminative-parity reading of group fairness: a system is fair if its rank-ordering of toxic from non-toxic comments is equally informative across identity subgroups. The stakeholder this metric protects is the identity-group member harmed when the model’s score function works less well for their group than for others, regardless of the direction the error tends. Following Borkan et al. [2019], we report three AUC variants per subgroup: *Subgroup AUC* (separability of toxic from non-toxic *within* the subgroup); *BPSN AUC*, in which the positive class is toxic comments not mentioning the identity and the negative class is non-toxic comments mentioning the identity (a low BPSN flags over-flagging of benign mentions); and *BNSP AUC*, in which the positive class is toxic comments mentioning the identity and the negative class is non-toxic background (a low BNSP flags under-flagging of toxicity targeting the group).

Identity	<i>n</i>	Subgroup AUC	BPSN AUC	BNSP AUC
male	2112	0.9367	0.9621	0.9592
female	2602	0.9418	0.9722	0.9515
homosexual gay or lesbian	538	0.8541	0.9298	0.9504
christian	2109	0.9475	0.9761	0.9471
jewish	411	0.9098	0.9541	0.9518
muslim	1054	0.8723	0.9424	0.9489
black	761	0.8670	0.9258	0.9573
white	1178	0.8738	0.9243	0.9593
psychiatric or mental illness	238	0.9560	0.9514	0.9753
<i>power-mean (p=−5) across the 9</i>		0.9018	0.9476	0.9554

Table 1: Per-identity bias AUCs on the private test set ($n = 97,320$). Identity counts use the 0.5 mention-threshold convention. Means agree with the competition-format header reported in cell1029.

Three asymmetries in Table 1 matter for the audit. First, the worst Subgroup AUCs cluster on identities with strong spurious training-time correlations to toxic examples: **homosexual_gay_or_lesbian** (0.854), **black** (0.867), **muslim** (0.872), and **white** (0.874). Second, BPSN ranks **black** and **white** lowest (0.926 and 0.924), the canonical “race-mention is toxicity-adjacent” pattern. Third, BNSP ranks **christian** lowest (0.947) and **muslim** second-lowest (0.949), indicating that toxicity *targeting* those groups is harder for the model to separate from non-toxic background; for **muslim**, the still-lower BPSN (0.942) means the BPSN-vs-BNSP rule classifies the error direction as over-flagging rather than under-flagging.

Combining BPSN and BNSP gives a direction of error per identity (full classification in `derived_direction_of_error`). Five identities are over-flagged: **white**, **black**, **psychiatric_or_mental_illness**, **homosexual_gay_or_lesbian**, **muslim**. Two are under-flagged: **christian**, **female**. The remaining two (**male**, **jewish**) lie within a ± 0.005 BPSN–BNSP balanced band (full table in `derived_direction_of_error.md`). The over- and under-flagged groups are disjoint, which sets up the stakeholder asymmetry we develop in §5.2: false-positive harms (silencing of identity mentions) and false-negative harms (toleration of targeted toxicity) accrue to different communities, and optimizing one fairness condition can worsen another (§5.2).

5.2 Fairness analysis with stakeholder-anchored metric choice

With Civil Comments base rates differing markedly across the nine evaluated identity subgroups (§3.1: $p \in [0.10, 0.32]$), the Chouldechova impossibility result [Chouldechova, 2017] states that the audited system *cannot* simultaneously satisfy predictive parity (equal PPV across groups) and error-rate balance

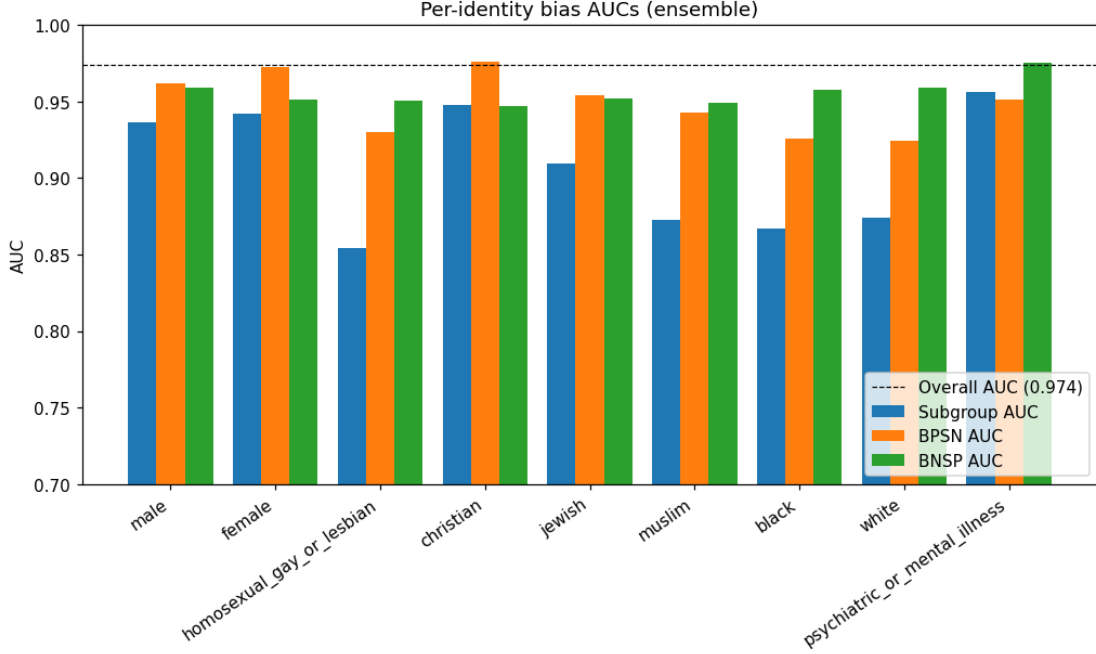


Figure 1: Visual rendering of Table 1: Subgroup, BPSN, and BNSP AUCs per evaluated identity. Low BPSN on **white** and **black** (race-mention over-flagging) and low BNSP on **christian** (under-flagging of toxicity targeting the group) drive the direction-of-error split. **muslim** has both metrics lowered; the BPSN-vs-BNSP rule classifies it as over-flagged.

(equal FPR and FNR across groups) unless base rates are equal (they are not; §3.1) or the classifier is perfect. The choice between these two normative readings of group fairness is therefore a stakeholder choice, not a technical detail. Predictive parity privileges the moderator: each flagged comment is equally likely to be genuinely toxic, regardless of which identity the comment mentions. Error-rate balance privileges the commenter: a non-toxic commenter who mentions any identity has equal odds of being wrongly silenced, and a victim of targeted toxicity has equal odds of being wrongly let through. We report both, alongside selection rate (statistical parity) and TPR (equal opportunity, Hardt et al., 2016), so neither choice is hidden from the reader.

We threshold the toxicity score at $s_{HR} = 0.5$ (motivated in §3.2); the impossibility result holds at any non-trivial threshold given the base-rate spread above.

Numerical demonstration of the trade-off. Take OVERALL PPV (0.6873) as a notional predictive parity target and ask, for each subgroup, what FPR would the model need to satisfy that target given its measured base rate and FNR? Holding base rate and FNR fixed and applying Chouldechova’s identity yields a required FPR (as an algebraic diagnostic, not an implementable adjustment) that ranges from 0.020 (**christian**) to 0.100 (**black**)—a $4.9\times$ ratio. Conversely, taking OVERALL FPR (0.0291) as an error-rate-balance target forces required PPV to span $[0.606, 0.883]$ —a 0.28 absolute spread. Either way, the trade-off between the two normative readings is substantial in this dataset, not a merely formal incompatibility (full computation in `derived_impossibility_demo.md`).

Stakeholder reading. The over- and under-flagged identities are disjoint (§5.1). Predictive parity, by holding PPV equal, would force FPR *up* on under-flagged identities (**christian**, **female**) and *down* on over-flagged identities; error-rate balance would do the reverse on PPV. No single member of the predictive-parity / error-rate-balance pair could have been optimized in a way that protects

Identity	n	base rate	sel. rate	PPV	TPR	FPR	FNR
OVERALL	97320	0.0799	0.0856	0.6873	0.7363	0.0291	0.2637
male	2112	0.1515	0.1061	0.7679	0.5375	0.0290	0.4625
female	2602	0.1345	0.0796	0.8406	0.4971	0.0147	0.5029
homosexual gay or lesbian	538	0.2862	0.1468	0.7975	0.4091	0.0417	0.5909
christian	2109	0.0996	0.0512	0.7870	0.4048	0.0121	0.5952
jewish	411	0.1655	0.0998	0.7317	0.4412	0.0321	0.5588
muslim	1054	0.2211	0.1195	0.7698	0.4163	0.0353	0.5837
black	761	0.3233	0.1840	0.8071	0.4593	0.0524	0.5407
white	1178	0.2997	0.1689	0.8040	0.4533	0.0473	0.5467
psychiatric or mental illness	238	0.2269	0.2059	0.7755	0.7037	0.0598	0.2963

Table 2: Per-subgroup confusion-matrix metrics at $s_{HR} = 0.5$. The competition’s bias-aware metric does not directly enforce predictive parity (equal PPV) or error-rate balance (equal FPR and FNR) across groups; this table makes the resulting violations visible. Source: `inference_outputs/cell1031`.

both stakeholders—the moderator who wants flagged-comment quality and the commenter who wants procedurally fair silencing/toleration odds—given the data the system was trained on. The audit’s contribution here is to make the choice visible, since the competition’s bias-aware metric (§5.3.2) is built on AUC-style measures and does not directly enforce either group-fairness condition.

5.3 Additional audit methods

5.3.1 Counterfactual robustness

Counterfactual fairness [Kusner et al., 2017] and the but-for-causation standard articulated in *Bostock v. Clayton County* [Supreme Court of the United States, 2020] both ask the same question: would the prediction change if *only* the protected attribute changed? Garg et al. [2019] adapt this framework to text classification by perturbing identity tokens within a fixed template sentence. We follow that operationalization for the toxicity classifier: twelve identity tokens substituted into ten templates spanning neutral and toxic registers, evaluated on all seven prediction heads of the BERT, LSTM, and ensemble models.

Operationalizing the do-intervention. Figure 2 makes a non-trivial methodological commitment visible. In conventional causal-fairness settings the protected attribute A is an isolated upstream variable and $\text{do}(A := a')$ has a well-defined meaning. In toxicity classification, A is a substring of the comment text C , and the “intervention” is implemented by editing C itself. Whether template substitution is a valid do-intervention on A depends on whether substituting different identity tokens yields equally fluent sentences: if “I am a Muslim” is grammatical but “I am a homosexual gay or lesbian” is not, any score gap between them is confounded by template-fluency artifacts.

Substitution-validity test. We test the do-intervention assumption empirically (cell1040). We score every counterfactual sentence under a pretrained GPT-2 language model and treat per-identity mean perplexity as a proxy for substitution fluency. Across the 120 counterfactual sentences, mean perplexity is 110.0 and the top-decile threshold is 214.9; the twelve flagged sentences cluster on identity strings that are themselves noun phrases (`psychiatric_or_mental_illness`, identity-level mean PPL 158.6) or stylistically marked (`heterosexual`, 136.5). This is the failure mode the plan flagged: a template that assumes a single-noun identity slot produces ungrammatical strings for compound identities, and the resulting score gaps for those identities cannot be cleanly attributed to identity effect alone. We

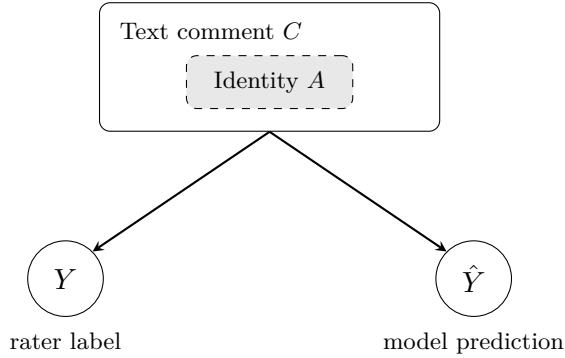


Figure 2: Anti-causal DAG for toxicity classification. The text comment C causes the human-rater label Y (the writer composes text; raters then read and judge), and the model predictor $\hat{Y}$ is also computed from C . The identity attribute A is not a separate upstream variable: it is a substring inside C (shaded box). Counterfactual interventions on A are operationalized in our experiments via template substitution within C , which conflates $\text{do}(A := a')$ with edits to the surrounding text and motivates the substitution-validity test in §5.3.1.

carry the flagged identities forward into the analysis but treat their counterfactual gaps as mixed signals of identity effect and substitution noise.

Counterfactual gaps. On neutral templates (e.g. “I am a [identity]”), the largest counterfactual scores fall on `black`, `homosexual_gay_or_lesbian`, `psychiatric_or_mental_illness`, `muslim`, and `white`—the same identities the Pillar 1 BPSN analysis flagged as over-flagged (§5.1), so counterfactual robustness corroborates rather than independently establishes the bias. The BERT and LSTM components have visibly different bias profiles per identity: on the toxicity head, `psychiatric_or_mental_illness` is heavily over-flagged by BERT (0.335) but lightly by LSTM (0.048); `asian` is heavily over-flagged by LSTM (0.350) but lightly by BERT (0.012). The Optuna blend (§4.3) places near-all weight on BERT, so the ensemble tracks BERT’s profile rather than averaging the two—a finding we return to in §6.4. Full heatmaps, marginal effects, and pairwise gap matrices live in `inference_outputs/cell04{0,1,2,3}`.

5.3.2 Critique of the competition metric

The bias-aware score the competition rewarded combines an Overall AUC on the full test set with a per-identity bias component,

$$\text{final} = w_{\text{overall}} \cdot \text{AUC}_{\text{overall}} + (1 - w_{\text{overall}}) \cdot M_p(\{\text{Subgroup}, \text{BPSN}, \text{BNSP}\}),$$

where M_p is the generalized power mean over the per-identity bias AUCs, and the competition fixed $w_{\text{overall}} = 0.25$ and $p = -5$. Each constant encodes a distinct normative claim.

Power mean $p = -5$ as difference-principle commitment. As $p \rightarrow -\infty$, M_p collapses toward the worst-performing subgroup, so the choice $p = -5$ embeds a difference-principle-style commitment to the worst-off subgroup — a narrower reading than Rawlsian fair equality of opportunity, which concerns pre-competition development rather than outcome equalization. Empirically, however, the bite of this choice is mild: sweeping $p \in \{1, 0.5, -1, -3, -5, -8, -15\}$ (`cell1044`) moves the final score by only ~ 0.004 , because the per-identity bias AUCs already lie in a tight band. The difference-principle framing is morally substantive but quantitatively mild on this model.

The 25/75 weighting. The choice $w_{\text{overall}} = 0.25$ is, by contrast, heavily binding. Sweeping w_{overall} from 0 to 1 (`cell1045/cell1046`) moves the final score from 0.9350 (all-bias) to 0.9737 (all-overall-AUC)—a 0.04 swing. At the chosen 0.25, the bias component does roughly three-quarters of the work in pulling the model’s headline number down from its Overall AUC. A different weighting, even 0.5, would have produced a different leaderboard and a structurally different “winning” solution.

Identity inclusion and exclusion. The metric averages bias AUCs over nine of the twenty-four identity columns Civil Comments collected (§3.1; `derived_identity_inclusion.md`). The cutoff appears to be sample-size driven: at the 0.5 mention threshold, all nine evaluated identities have $n \geq 238$ and all fifteen excluded identities have $n \leq 217$. The operational rule is defensible; its normative consequence is that the harms experienced by under-represented identities—`transgender`, `bisexual`, `asian`, `latino`, `hindu`, `buddhist`, `atheist`, and every disability category other than `psychiatric_or_mental_illness`—do not enter the score the competition rewarded. “We evaluate where we have data” compounds the visibility problem of marginalized identities: the groups most likely to suffer rare-but-severe harms are precisely the ones the metric cannot see.

Together, these three constants—Rawlsian aggregation, the fairness–utility weighting, and the identity-inclusion cutoff—encode distinct normative commitments the competition metric ships without flagging [Stoyanovich and Howe, 2019, Brown et al., 2021]. The audit renders them visible so participants can decide whether the metric’s defaults match the fairness theory they want to defend.

6 Summary

6.1 Was the data appropriate?

Civil Comments is operationally usable but rests on a contested measurement: toxicity is a fraction-of-rater agreement, binarized at 0.5, on a rater pool that is not demographically balanced to the populations the labels concern (§3.3). The dataset’s identity-coverage spans twenty-four columns, but the bias-aware metric we audit against evaluates only nine of them (§5.3.2). We accept the data as appropriate for the audit’s narrow goal of reproducing and stress-testing the third-place solution, but emphasize that every fairness conclusion below inherits the upstream measurement choice: a different aggregation rule or a demographically rebalanced rater pool could move some of the gaps reported here materially.

6.2 Robustness, accuracy, and fairness assessment

The implementation is accurate on the metric the competition rewarded (Overall AUC 0.97368, final score 0.94464). It is not robust or fair under either of the two stakeholder-grounded readings of group fairness we tested (predictive parity and error-rate balance). Pillar 1 flags five over-flag and two under-flag identities, with error-direction falling on disjoint groups (§5.1). Pillar 3 confirms substantial dispersion in PPV ([0.73, 0.84]), FPR ([0.012, 0.060]), and selection rate (§5.2), and Chouldechova’s impossibility result rules out simultaneous predictive parity and error-rate balance at the observed base rates—forcing a stakeholder choice the developers made implicitly through their training objective. Pillar 2’s counterfactual gaps fall on the same identities Pillar 1 flagged, providing independent corroboration that the bias is real rather than a test-set artifact.

We privilege the commenter as the stakeholder of greatest concern: false-positive silencing of identity-mentioning speech is concentrated on minority sexual orientations, racial groups, and mental-health mentions (§5.1), so error-rate balance is the fairness condition whose violation we treat as most consequential. Under that anchor, the implementation is unfair: FPR varies fivefold across the nine evaluated identities at $s_{HR} = 0.5$.

6.3 Public-sector or industry deployment?

Public sector: no. The differential silencing of identity-mentioning speech (§5.1, §6.2) would raise speech-governance and disparate-impact concerns, and the toxicity label’s measurement choice (whose toxicity, rated by whom) is incompatible with the neutrality typically demanded of government speech regulation. **Industry, with guardrails: perhaps.** A defensible deployment would require at minimum (a) threshold-sensitivity reporting and human-review escalation for high-cost error modes, with any per-identity-direction tuning embedded in a stakeholder-aware operating-point process rather than silently inheriting a single s_{HR} ; (b) continuous monitoring of disparate impact across the full twenty-four identity columns, not just the nine; (c) auditing of the rater pool’s demographic composition relative to the user population; and (d) a public model card [Mitchell et al., 2019] and dataset/model nutrition label [Stoyanovich and Howe, 2019] documenting the metric’s value choices and the impossibility-result trade-off. Without these, the system as audited is closer to a research artifact than a deployment-ready ADS.

6.4 Recommendations

Data and labels. Retain rater-level demographics so labels can be disaggregated by annotator identity; the current aggregate fraction collapses information needed to evaluate whose toxicity standard the system enacts (§3.3). Report downstream metrics at multiple thresholds (e.g., 0.25, 0.5, 0.75) rather than one—binarizing at 0.5 embeds a majoritarian decision rule that minority readings of toxicity cannot push back against.

Methodology. Expand per-identity evaluation beyond the nine sample-size-cutoff columns to all twenty-four, with confidence bounds for smaller-sample identities (§5.3.2). Augment the bias-aware AUC objective with a Module 1 metric (e.g., per-group PPV or FPR variance) at training time, so the metric’s normative defaults are not inherited unexamined [Raji et al., 2020]. Run counterfactual substitution tests with the GPT-2 perplexity filter of §5.3.1 inside validation, not as a post-hoc audit. If structural diversity is the fairness mechanism the original F.H.S.D.Y. ensemble exploited, restore its missing components before further blend-weight tuning—the BERT-dominated blend here ($w_{\text{BERT}} = 0.993$) may amplify a single component’s inductive bias rather than averaging across components, as Pillar 2’s BERT-vs-LSTM divergence suggests.

6.5 Limitations

Five limitations bound the conclusions above. (i) The audit covers a BERT-large + LSTM-GRU recreation, not the full five-component F.H.S.D.Y. ensemble (§4.1); structural-diversity contributions of the missing components are absent from the bias profile. (ii) The toxicity target is itself a fraction-of-rater measurement binarized at 0.5, with rater-demographic validity contestable (§3.3). (iii) Twelve counterfactual substitutions are fluency-degraded (§5.3.1); their gaps mix identity effect with template artifact. (iv) Per-subgroup confusion-matrix and counterfactual marginals are point estimates without bootstrap CIs, weakening interpretive weight on small- n subgroups (`psychiatric_or_mental_illness`, $n = 238$; `homosexual_gay_or_lesbian`, $n = 538$). (v) The audit is column-wise; intersectional harms for comments mentioning multiple identity groups are not estimated.

7 Project Contributions

Both authors contributed equally to all parts of this project: the recreation of the BERT-large + LSTM-GRU ensemble and the Optuna-blended inference pipeline (§4); the design and execution of

the bias-AUC, Module 1 fairness, counterfactual robustness, and metric-critique experiments (§5); the visualization layer; and the report prose. No work was siloed by team member.

References

- Daniel Borkan, Lucas Dixon, Jeffrey Sorensen, Nithum Thain, and Lucy Vasserman. Nuanced metrics for measuring unintended bias with real data for text classification. In *Companion Proceedings of The 2019 World Wide Web Conference (WWW '19 Companion)*. ACM, 2019. doi: 10.1145/3308560.3317593.
- Shea Brown, Jovana Davidovic, and Ali Hasan. The algorithm audit: Scoring the algorithms that score us. *Big Data & Society*, 8(1), 2021. doi: 10.1177/2053951720983865.
- Alexandra Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163, 2017. doi: 10.1089/big.2016.0047.
- Sahaj Garg, Vincent Perot, Nicole Limtiaco, Ankur Taly, Ed H. Chi, and Alex Beutel. Counterfactual fairness in text classification through robustness. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society (AIES '19)*. ACM, 2019. doi: 10.1145/3306618.3317950.
- Moritz Hardt, Eric Price, and Nathan Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems 29 (NeurIPS 2016)*, 2016.
- Matt J. Kusner, Joshua R. Loftus, Chris Russell, and Ricardo Silva. Counterfactual fairness. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017.
- Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*. ACM, 2019. doi: 10.1145/3287560.3287596.
- Inioluwa Deborah Raji, Andrew Smart, Rebecca N. White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson, Jamila Smith-Loud, Daniel Theron, and Parker Barnes. Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT* '20)*. ACM, 2020. doi: 10.1145/3351095.3372873.
- sakami0000. 3rd-place solution to the jigsaw unintended bias in toxicity classification kaggle competition. Kaggle competition entry, 2019.
- Julia Stoyanovich and Bill Howe. Nutritional labels for data and models. *IEEE Data Engineering Bulletin*, 42(3), 2019.
- Supreme Court of the United States. *Bostock v. clayton county, georgia*. 590 U.S. ____, 140 S. Ct. 1731, 2020.